

POPULAR OPEN SOURCE SOFTWARE ACROSS ARCHITECTURE LAYERS

Category	Software	Description
Frontend	React	Developed by Facebook, known for its component-based architecture, virtual DOM for efficient updates, and vast ecosystem. Ideal for building highly interactive and dynamic interfaces.
	Angular	A comprehensive framework by Google, based on TypeScript, with a strong focus on maintainability and scalability. Well-suited for large-scale enterprise applications.
	Vue.js	A progressive framework known for its flexibility, ease of learning, and high performance. Often used for building single-page applications (SPAs) and integrating with existing projects.
Backend	Node.js	A JavaScript runtime environment, enabling you to use JavaScript on the server side. Known for its event-driven, non-blocking I/O model, making it efficient for building real-time applications and APIs.
	Django	A high-level Python framework, known for its “batteries-included” philosophy, providing many built-in features for rapid development. Promotes clean, pragmatic design and follows the model-view-template (MVT).
	Spring Boot	A Java framework that simplifies the bootstrapping and development of Spring applications. Well-suited for building enterprise-grade Java applications, offering a wide range of features and modules.
	Ruby on Rails	A popular web framework built on Ruby, known for its “convention over configuration” approach, which reduces the amount of configuration needed. Emphasises developer happiness and productivity.
Database	PostgreSQL	A powerful, open source relational database management system (RDBMS), known for its reliability, data integrity, and extensibility. Supports advanced features like complex queries, transactions, and triggers.
	MySQL	A widely used open source RDBMS, known for its ease of use, speed, and large community support. Commonly used for web applications and content management systems.
	MongoDB	A popular NoSQL database, designed for flexible document-based data storage. Offers high scalability, performance, and availability, making it a good choice for applications with rapidly changing data models.
Web server	Nginx	A high-performance web server and reverse proxy, known for its lightweight design, efficient resource utilisation, and ability to handle a large number of concurrent connections. Often used for load balancing and caching.
	Apache	One of the most popular web servers globally, offering a wide range of features and modules. Known for its flexibility and stability, but may consume more resources than Nginx under heavy loads.
Cloud infra	Kubernetes	An open source container orchestration platform, automating the deployment, scaling, and management of containerised applications. Offers high availability, load balancing, and self-healing capabilities.
	Docker	A platform for building, shipping, and running applications in containers. Containers provide a consistent runtime environment for applications, making them portable across different environments.
DevOps	Jenkins	An open source automation server, enabling continuous integration and continuous delivery (CI/CD) pipelines. Supports a wide range of plugins and integrations for building, testing, and deploying applications.
	GitLab CI/CD	A built-in CI/CD platform within GitLab, allowing you to automate the entire software delivery process within a single application.
	Ansible	An open source IT automation tool used for configuration management, application deployment, and task automation. Employs a simple, agent-less architecture and uses YAML for defining automation tasks.

transformed the way modern applications are deployed and managed. Cloud infrastructure provides on-demand computing resources, allowing for scalability and cost-efficiency by paying only for the resources used. Containerisation, on the other hand, packages applications and their dependencies into isolated units called containers, ensuring consistent behaviour across different environments.

Kubernetes, a leading container orchestration platform, automates the deployment, scaling, and management of containerised applications, making it easier to achieve high availability, fault tolerance, and efficient resource utilisation.

Docker, a popular containerisation platform, simplifies the process of creating, sharing, and running containers, promoting portability and reproducibility of applications.

The combination of cloud infrastructure and containerisation empowers enterprises to build agile, scalable, and resilient applications that can adapt to changing demands and optimise resource allocation.

DevOps tools

DevOps tools bridge the gap between development and operations teams, fostering collaboration, automating processes, and accelerating the software delivery lifecycle. They facilitate continuous integration (CI) and continuous delivery (CD), enabling faster feedback loops, quicker bug fixes, and more frequent releases. Jenkins, a widely adopted automation server, provides a comprehensive framework for building CI/CD pipelines, orchestrating various stages of the software development process, and integrating with numerous tools and platforms.

GitLab CI/CD, integrated within the GitLab platform, offers a unified solution for code management, version control, and automated